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Roll No

ME-701(B)-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Computer Aided Engineering and FEM

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) What are skeletal and continuum structures.
 - b) How modelling for infinite DOF is done?
2.
 - a) Explain the term Shape function.
 - b) Why polynomial terms are preferred for shape functions in finite element method?
3.
 - a) What is Boundary conditions? Give examples.
 - b) State different methods used for the solution of simultaneous equations.
4. Discuss 2D rectangular and triangular elements. What are axisymmetric elements?
5. Explain the following briefly:
 - a) Effect of element aspect ratio on accuracy
 - b) Numbering nodes for bandwidth minimization

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6. Using variational approach derive element stiffness matrix of.
 - a) Bar element
 - b) Plane truss element
7.
 - a) Explain the different commercial software packages used for static analysis of frames.
 - b) Discuss in details the solution of simultaneous equations using any one method.
8. Write short notes on followings:
 - a) Hamilton's principle
 - b) Consistent and lumped mass matrices
